

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50

Annexure A Data Interpretation

Code of Conduct:

*I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Q. No	Understanding of Data (25%)	Analysis (25%)	Application of Concepts (20%)	Conclusion (20%)	Clarity (10%)	Total Score (100%)
1						
2						
3						
4						
5						
Total Marks (Out of 100)					Total Marks (Out of 50)	

Annexure A

Data Interpretation

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method Normal Occlusion Low Light

Viola-Jones 80% 60% 55%

YOLO 90% 75% 70%

Deep Learning 95% 85% 80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

Answer:

Given Data

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Understanding of Data

The data shows that all three methods perform best under normal conditions and their accuracy decreases when **occlusion** and **low-light conditions** are introduced. Deep Learning maintains the highest accuracy in all three environments. The given data and comparison requirement are specified in the assessment.

CO4_AT1_Data Interpretation

Analysis

Viola-Jones

- Normal: **80%**
- Occlusion: **60%**
- Low light: **55%**
- Accuracy decreases significantly under difficult environmental conditions.

- It has the lowest performance among the three methods.

YOLO

- Normal: **90%**
- Occlusion: **75%**
- Low light: **70%**
- Performs considerably better than Viola-Jones.
- Shows better robustness to environmental changes.

Deep Learning

- Normal: **95%**
- Occlusion: **85%**
- Low light: **80%**
- Achieves the highest accuracy in every condition.
- Its performance decreases less severely compared with the other methods.

Overall Comparison

Method	Average Accuracy
Viola-Jones	65.0%
YOLO	78.3%
Deep Learning	86.7%

Deep Learning has the best overall performance.

Trade-offs

- **Viola-Jones:** Lower accuracy in challenging environments but relatively simple.
- **YOLO:** Provides a good balance between detection accuracy and practical object detection.
- **Deep Learning:** Provides the highest robustness and accuracy, but generally requires greater computational resources and training data.

Conclusion

Deep Learning is the best technique for robust detection across varying environmental conditions. It achieves 95% accuracy in normal conditions, 85% under occlusion, and 80% in low light. Therefore, for applications such as surveillance, autonomous systems, and real-world object detection, Deep Learning is the most suitable choice.

2. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions

- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

Answer:

Method A

- Produces **stable and reliable motion vectors**.
- Less affected by noise.
- May fail to detect small or subtle movements.
- Suitable when stable overall motion estimation is more important than fine details.

Method B

- Detects **small and fine movements** effectively.
- More sensitive to noise.
- Noisy vectors can cause incorrect motion estimation.
- Suitable when detecting subtle movements is critical.

Comparison

Factor	Method A	Method B
Vector stability	High	Low
Small-motion detection	Low	High
Noise sensitivity	Low	High
Reliability	High	Moderate
Fine-motion analysis	Moderate/Low	High

Application-Based Evaluation

For a **dynamic scene**, the choice depends on the application.

If the objective is **stable tracking of large objects**, Method A is preferable because stable vectors reduce false motion information.

If the objective is **detecting small and rapid movements**, Method B is more appropriate because it can capture fine motion.

However, Method B should preferably be combined with **noise filtering or smoothing** before tracking.

Conclusion

For general dynamic-scene motion tracking, **Method A is more reliable** because stable

vectors provide consistent tracking. However, **Method B is better for applications requiring fine-motion detection.** Therefore, the final choice should depend on whether stability or sensitivity to small motion is the primary requirement.

3. Background Modeling Strategy (GMM)

A GMM-based system shows:

- High false positives in dynamic background
- Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

Answer:

Understanding of Data

A **Gaussian Mixture Model (GMM)** represents the background using multiple Gaussian distributions. It can model different pixel intensity patterns, but its performance can be affected by changes in the scene.

Dynamic Background

In a dynamic background, objects such as:

- Moving trees
- Water
- Shadows
- Moving curtains

may be incorrectly identified as foreground.

This results in **high false positives**.

Static Background

In a static background, the system may fail to detect some changes if the model adapts incorrectly or if foreground objects become incorporated into the background model.

This results in **missed detections**.

Comparison

Scenario	GMM Performance	Main Problem
Dynamic background	Poor	High false positives
Static background	Better but may miss changes	Missed detections

Is GMM Alone Sufficient?

No. GMM alone is not sufficient for reliable background subtraction in all environments.

A robust system should combine GMM with additional techniques.

Proposed Improvements

1. **Adaptive learning rate**
Adjust the background model according to scene changes.
2. **Morphological filtering**
Remove small noise and fill gaps in detected objects.
3. **Shadow detection**
Prevent shadows from being classified as foreground objects.
4. **Temporal filtering**
Use information from consecutive frames to confirm actual movement.
5. **Object tracking**
Track detected objects across frames to reduce false detections.
6. **Deep-learning-based detection**
A CNN/object detector can be used together with background subtraction for improved accuracy.

Conclusion

GMM is useful for background modeling but **should not be used alone** when the environment contains complex motion or changing backgrounds. Combining GMM with adaptive modeling, filtering, shadow removal, temporal analysis, and object detection can significantly improve reliability.

4. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

Answer:

Comparison

Factor	Method A	Method B
Error	5%	10%
Accuracy	Higher	Moderate
Computational cost	High	Low
Speed	Lower	Higher
Real-time suitability	Moderate	High
Offline suitability	High	Moderate

Method A

Method A provides only **5% error**, making it more accurate.

However, its high computational requirement can increase processing time. Therefore, it is

more suitable when accuracy is more important than speed.

Best application: Offline video analysis, forensic analysis, high-precision motion estimation.

Method B

Method B has **10% error**, which is higher than Method A.

However, its low computational requirement makes it faster and more suitable for systems with limited processing resources.

Best application: Real-time surveillance, robotics, interactive systems, and embedded devices.

Real-Time vs Offline Decision

Real-time applications → Method B

Real-time systems require quick processing. Accepting a moderate increase in error can be reasonable when low latency is essential.

Offline applications → Method A

Offline applications can tolerate higher computational time because the main objective is obtaining more accurate results.

Conclusion

Method A is preferred for offline applications because of its lower error rate, while Method B is preferred for real-time applications because of its low computational cost. Therefore, the best method depends on the application requirement rather than accuracy alone.

5. Object Detection under Scale Variation

Results:

Object Size Accuracy

Large 94%

Medium 80%

Small 55%

Compare detection performance and evaluate system limitations. Justify improvements.

Answer:

Given Data

Object Size	Accuracy
Large	94%
Medium	80%
Small	55%

The assessment specifically asks for comparison of detection performance and evaluation of system limitations.

CO4_AT1_Data Interpretation

Understanding of Data

The system performs very well when objects are **large**, but accuracy decreases as object size becomes smaller.

The reduction is:

- Large → Medium: **14 percentage points**
- Medium → Small: **25 percentage points**
- Large → Small: **39 percentage points**

This clearly shows that **small-object detection is the major limitation**.

Analysis

Large objects – 94%

- Very high detection accuracy.
- Objects contain sufficient visual information.
- Features can be extracted more easily.

Medium objects – 80%

- Accuracy decreases by 14 percentage points.
- Some object details become less distinguishable.

Small objects – 55%

- Lowest accuracy.
- Important visual features occupy fewer pixels.
- Objects are more easily affected by image noise, resolution, and occlusion.

System Limitation

The major limitation is **poor detection of small objects**.

The difference between large and small object accuracy is:

$$94\% - 55\% = 39 \text{ percentage points}$$

This is a substantial performance reduction.

Proposed Improvements

1. **Higher-resolution input images**
Preserve more details of small objects.
2. **Multi-scale feature extraction**
Extract features at different image scales.
3. **Feature Pyramid Networks (FPN)**
Improve representation of objects at different sizes.
4. **Image tiling**
Divide large images into smaller regions so that small objects occupy more pixels.
5. **Data augmentation**
Include different scales and transformations during training.
6. **Small-object-focused training data**
Increase the number and diversity of small-object examples.

Conclusion

The system performs excellently for large objects with **94% accuracy**, but its performance falls to **55% for small objects**. Therefore, scale variation is a significant limitation. Multi-scale feature extraction, higher-resolution images, image tiling, and appropriate training augmentation can improve small-object detection.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application ; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

